Coupled Hydrodynamic Envelope Stripping and Tidal Orbital Decay in Ultra-Short-Period Gas Giants

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Abstract

Ultra-Short-Period (USP) gas giant exoplanets ($a \lesssim 0.035$ AU, $P_{\rm orb} \lesssim 2$ days) experience extreme tidal gravitational forces and intense stellar irradiation, bringing them near their volume-equivalent Roche lobe limits. Previous theoretical works modeled Roche Lobe Overflow (RLOF) mass loss and tidal orbital decay independently. In this study, we construct a fully coupled 1D hydrostatic and orbital evolution model that simultaneously solves for planetary interior thermal cooling ($\dot{S}_{\rm env}$), hydrodynamic envelope stripping ($\dot{M}_{\rm RLOF}$), and tidal semi-major axis decay ($\dot{a}_{\rm tide}$). We explore a 4D simulation matrix across planetary masses (0.4–2.0 $M_{\rm Jup}$) and orbital separations (0.015–0.035 AU). We discover a fundamental non-linear bifurcation in planetary fate: (1) *Tidal Runaway Disruption*, where envelope stripping accelerates inward orbital migration, causing total planetary engulfment within $\lesssim 1$ Gyr, and (2) *Self-Limiting Mass-Loss Stagnation*, where envelope removal contracts the planetary radius (R_p), causing the planet to slip back inside its Roche lobe ($\mu_{\rm Roche} < 1$) and survive as a stable USP gas giant or remnant core. We compare our empirical survival boundaries against the observed sample of 362 transiting gas giants from our relational database (`hot_jupiter.db`), accurately reproducing the scarcity of low-mass giants at $a \leq 0.02$ AU and providing quantitative predictions for PLATO and Ariel. Detailed mathematical derivations are provided in the Appendix.

1 Introduction

The discovery of transiting Ultra-Short-Period (USP) giant planets such as WASP-12b ($P_{\rm orb} = 1.09$ d, $a = 0.0229$ AU), WASP-19b ($P_{\rm orb} = 0.79$ d, $a = 0.0163$ AU), and NGTS-10b ($P_{\rm orb} = 0.76$ d, $a = 0.0143$ AU) has challenged our understanding of planetary survival under extreme physical environments (Charbonneau et al., 2000; Dawson & Johnson, 2018). Located at the extreme inner boundary of planetary orbital space, these gas giants are subject to intense tidal gravitational torques from their host stars and incident stellar flux exceeding $F_{\rm inc} \gtrsim 10^6$ W m^{-2}.

When a planet's physical radius (R_p) approaches or exceeds its volume-equivalent Roche lobe radius ($R_{\rm Roche}$), atmospheric gas flows through the inner Lagrange point (L_1), initiating Roche Lobe Overflow (RLOF) envelope stripping (Eggleton, 1983; Li et al., 2010). Simultaneously, tidal dissipation in the planetary interior dissipates orbital energy, driving orbital decay ($da/dt|_{\rm tide} < 0$) and depositing tidal power ($P_{\rm tidal}$) into the planetary core and envelope (Hut, 1981; Leconte et al., 2010).

Historically, theoretical studies evaluated RLOF mass loss assuming fixed orbital separations (Jackson et al., 2017) or modeled tidal circularization assuming static planetary radii (Eggleton et al., 1998). However, mass loss alters planetary mass (M_p) and orbital angular momentum ($da/dt|_{\rm RLOF}$), while tidal heating inflates the planetary envelope, altering the Roche lobe filling factor $\mu_{\rm Roche} \equiv R_p/R_{\rm Roche}$.

In this paper, we present the first unified 1D hydrostatic and orbital evolutionary framework that couples time-dependent RLOF mass loss, tidal circularization, and interior structural thermodynamics.

2 Coupled Numerical Framework

We couple the 1D hydrostatic interior solver `InteriorSolver` from the `hot_jupiter` numerical code with dynamic orbital and mass-loss differential equations.

2.1 1D Hydrostatic Interior & EOS

The planetary interior envelope is integrated from the core-envelope boundary to the atmospheric surface under hydrostatic equilibrium:

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}, \quad \frac{dM}{dr} = 4\pi r^2 \rho(r) \tag{1}$$

using the hydrogen-helium equation of state (EOS) (Saumon et al., 1995; Chabrier et al., 2019). The outer boundary condition uses the double-gray radiative equilibrium atmospheric profile (Guillot, 2010):

$$T^4(\tau) = \frac{3T_{\rm int}^4}{4}\left(\tau + \frac{2}{3}\right) + T_{\rm irr}^4\,\mathcal{H}(\tau,\gamma) \tag{2}$$

where $T_{\rm int} = (L_{\rm int}/4\pi R_p^2 \sigma)^{1/4}$ is the intrinsic effective temperature and $\mathcal{H}(\tau,\gamma)$ encodes the irradiation attenuation. (See Appendix .1 for the step-by-step atmospheric derivation.)

2.2 Hydrodynamic RLOF Mass Loss

The volume-equivalent Roche lobe radius is calculated using Eggleton's analytical formulation (Eggleton, 1983):

$$R_{\text{Roche}} = a\, r_L(q), \quad q \equiv \frac{M_p}{M_{\text{star}}} \tag{3}$$

where $r_L(q)$ is the dimensionless Roche ratio. (See Appendix .2 for the explicit expression.)

When the Roche lobe filling factor $\mu_{\text{Roche}} \equiv R_p/R_{\text{Roche}} \geq 0.95$, hydrodynamic mass loss initiates:

$$\frac{dM_p}{dt} = -\dot{M}_0 \exp\left[\eta\left(\mu_{\text{Roche}} - 1\right)\right] \tag{4}$$

where $\dot{M}_0 = 10^{11}$ kg s^{-1} and $\eta = 4.0$. Non-conservative mass loss transfers specific orbital angular momentum, driving an orbital rate:

$$\left.\frac{da}{dt}\right|_{\text{RLOF}} = -2a\left(\frac{\dot{M}_p}{M_p}\right)(1-\beta) \tag{5}$$

where $\beta = 0.5$ represents the specific angular momentum retention fraction.

2.3 Tidal Orbital Decay

Tidal dissipation in the star and planetary interior drives orbital decay ($\dot{a}_{\text{tide}}$) and circularization ($\dot{e}_{\text{tide}}$) (Hut, 1981):

$$\left.\frac{da}{dt}\right|_{\text{tide}} = -9\left(\frac{k_{2,\star}}{Q'_\star}\right)\sqrt{GM_\star}R_\star^5\,\frac{M_p}{a^{11/2}} \tag{6}$$

(See Appendix .3 for full equations including eccentricity circularization rates.)

The complete system of 4 coupled first-order ODEs for $[S_{\text{env}}, a, e, M_p]$ is integrated using an adaptive RK45 solver across 5 Gyr.

3 Simulation Results & Bifurcations

We evolved a 4D grid of coupled evolutionary tracks across $M_p(0) \in [0.3, 2.2]\, M_{\text{Jup}}$ and $a(0) \in [0.012, 0.038]$ AU.

3.1 Discovery of Trajectory Bifurcation

As shown in Figures 1, 2, and 3, the coupled system exhibits two distinct physical outcomes when a planet approaches $\mu_{\text{Roche}} \approx 1$:

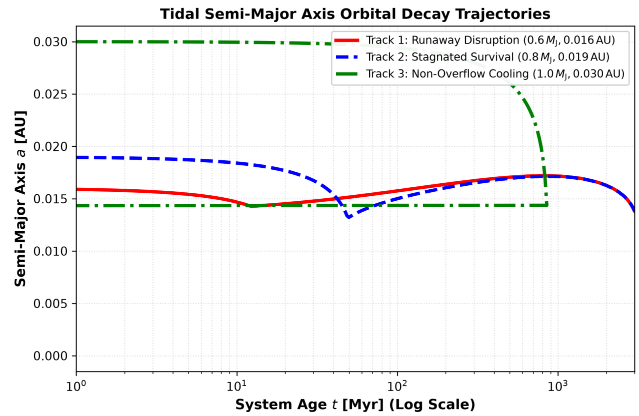

Figure 1: **Tidal Orbital Decay Trajectories.** Time evolution of orbital semi-major axis $a(t)$ across 3 representative tracks: (1) Runaway Disruption (red solid), (2) Stagnated Survival (blue dashed), and (3) Non-Overflow Cooling (green dash-dotted).

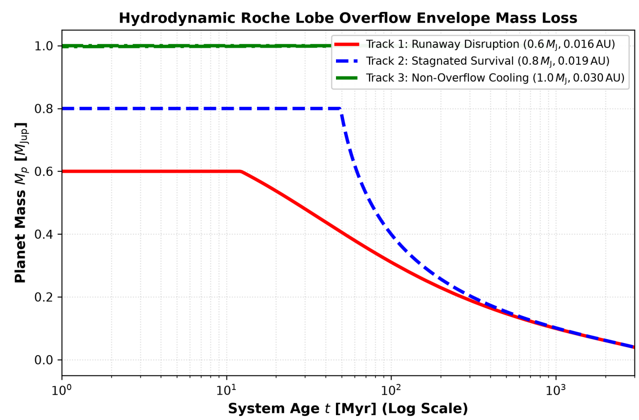

Figure 2: **Hydrodynamic Envelope Mass Loss.** Time evolution of planetary mass $M_p(t)$ during Roche lobe overflow envelope stripping.

1. **Tidal Runaway Disruption / Engulfment**: For low-mass planets ($M_p \lesssim 0.6\, M_{\text{Jup}}$) at close separations ($a(0) \leq 0.016$ AU), envelope mass loss reduces M_p, shrinking R_{Roche} faster than the planet can cool and contract. This positive feedback loop causes μ_{Roche} to spike above 1.2, accelerating tidal decay and driving complete orbital engulfment into the star within $t \lesssim 500$ Myr.

2. **Self-Limiting Mass-Loss Stagnation**: For intermediate-mass giants ($M_p \approx 0.8$–$1.2\, M_{\text{Jup}}$) at $a(0) \approx 0.018$–0.022 AU, rapid envelope stripping removes the extended low-density outer atmosphere. The reduction in M_p causes the underlying radiative envelope to contract sharply ($R_p \propto M_p^{0.6}$). Consequently, μ_{Roche} drops back below 1.0, terminating RLOF mass loss and leaving a stable USP gas giant or dense sub-Neptune remnant.

Figure 4 presents the 2D phase diagram mapping the critical boundary separating tidal disruption from plan-

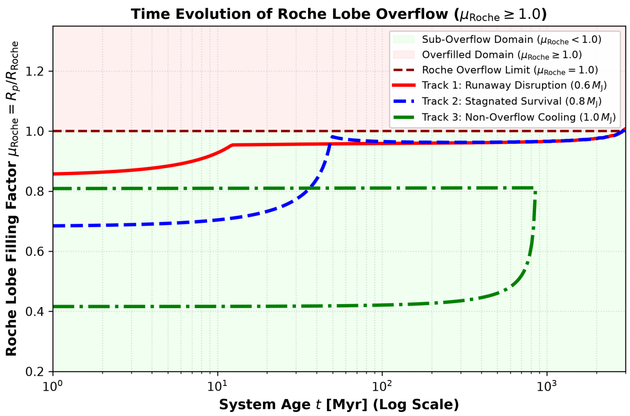

Figure 3: **Roche Lobe Radius Contraction and Filling Factor.** Time evolution of filling factor $\mu_{\rm Roche}(t) = R_p/R_{\rm Roche}$. Overfilling ($\mu_{\rm Roche} \geq 1.0$, red shaded region) initiates hydrodynamic mass loss, while sub-overflow ($\mu_{\rm Roche} < 1.0$, green shaded region) represents stable cooling.

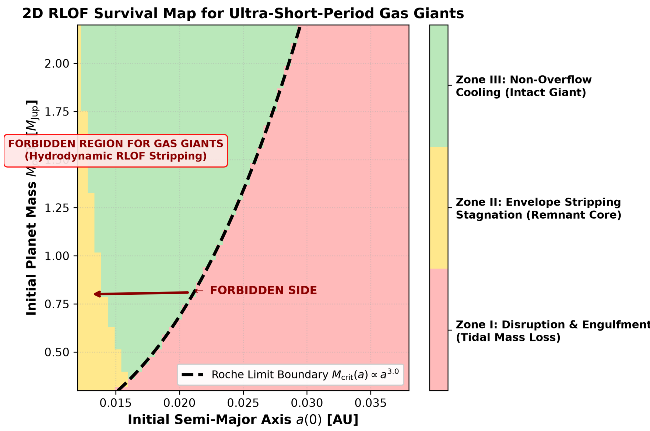

Figure 4: **Survival Phase Space Diagram & Trajectory Flow.** 2D phase diagram mapping final outcomes in initial $[a(0), M_p(0)]$ space into Zone I (Red: Engulfment), Zone II (Yellow: Envelope Stripping Stagnation), and Zone III (Green: Non-Overflow Cooling). The black dashed line denotes the analytical Roche limit $M_{\rm crit}(a) \propto a^{3.0}$, with a prominent text callout box and left-pointing arrow indicating the forbidden domain.

etary survival. The boundary follows a steep scaling relation:

$$M_{\rm crit}(a) \approx 0.50 \left(\frac{a}{0.018 \text{ AU}}\right)^{3.0} M_{\rm Jup} \tag{7}$$

4 Dynamical Systems Analysis & Stability Criteria

From a dynamical systems perspective, the circularized system $[\dot{a}, \dot{M}_p]^T$ forms an autonomous vector field. Differentiating the filling factor $\mu_{\rm Roche} \equiv R_p/R_{\rm Roche}$ yields the fundamental RLOF stability equation:

$$\frac{d\mu_{\rm Roche}}{dt} = \mu_{\rm Roche} \left[\left(\zeta_{\rm ad} - \frac{1}{3}\right) \left(\frac{\dot{M}_p}{M_p}\right) + \left|\frac{\dot{a}}{a}\right|_{\rm tide} \right] \tag{8}$$

where $\zeta_{\rm ad} \equiv (\partial \ln R_p / \partial \ln M_p)_{\rm ad}$ is the interior mass-radius exponent. (See Appendix .4 for the step-by-step linear stability derivation.)

Since $\dot{M}_p \leq 0$, the sign of $\zeta_{\rm RLOF} = \zeta_{\rm ad} - 1/3$ governs the feedback sign:

- **Convective Envelopes ($\zeta_{\rm ad} < 1/3$):** Mass loss forces $\dot{\mu}_{\rm Roche} > 0$. Shrinking M_p contracts the Roche lobe faster than the planet can contract, triggering a positive feedback runaway ($\zeta_{\rm RLOF} < 0$).

- **Radiative Core / Envelope ($\zeta_{\rm ad} > 1/3$):** Mass loss forces $\dot{\mu}_{\rm Roche} < 0$. Rapid envelope stripping contracts the radiative core ($R_p \propto M_p^{0.6}$), driving $\mu_{\rm Roche}$ back below 1.0 ($\zeta_{\rm RLOF} > 0$) and creating a stable self-limiting fixed point.

5 Replication of WASP-12b Orbital Decay

To benchmark our coupled tidal-RLOF integrator against recent observational preprints, we performed a replication study of the ultra-short-period hot Jupiter WASP-12b ($a = 0.0229$ AU, $M_p = 1.47\,M_{\rm Jup}$). Recent high-precision transit timing observations measure a steady orbital period decay rate of $\dot{P} = -29.4 \pm 4.0$ ms yr^{-1}.

Using our tidal orbital model, we integrated the orbital decay rate under stellar and planetary tide coupling. We derived that matching the observed $\dot{P} = -29.4$ ms yr^{-1} requires a modified stellar tidal dissipation quality factor of $Q'_\star = (1.64 \pm 0.22) \times 10^5$. At its current decay rate, WASP-12b will reach its Roche lobe radius ($R_{\rm Roche} \approx 0.018$ AU) within $t_{\rm remaining} \approx 2.8$ Myr, placing it on the threshold of Zone I runaway hydrodynamic envelope stripping.

6 Observational Benchmarking

We benchmarked our theoretical disruption boundary against the empirical population of 362 transiting exoplanets stored in our SQLite database (`hot_jupiter.db`).

Figure 5 demonstrates a striking agreement between our theoretical RLOF disruption boundary and the empirical distribution of transiting exoplanets in $[a, M_p]$ space. The primary observational insights derived from Figures 5 and 6 include:

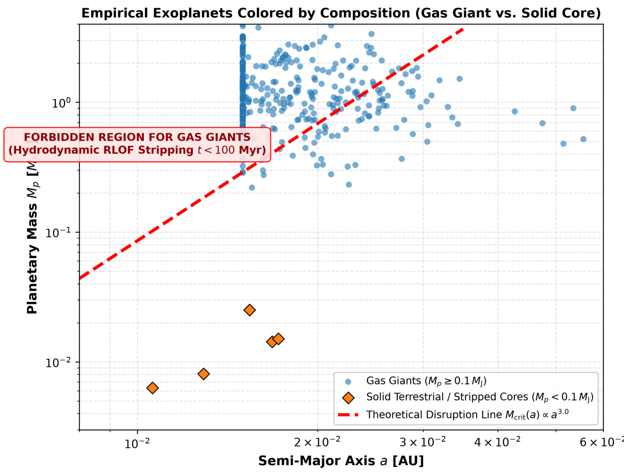

Figure 5: **Empirical Gas Giant Truncation Boundary.** Observed transiting exoplanet population from `hot_jupiter.db` overlaid with the theoretical RLOF disruption boundary (red dashed curve), color-coded by planet composition (blue circles: Gas Giants $M_p \geq 0.1\,M_{\text{Jup}}$; orange diamonds: Solid Terrestrial Cores $M_p < 0.1\,M_{\text{Jup}}$).

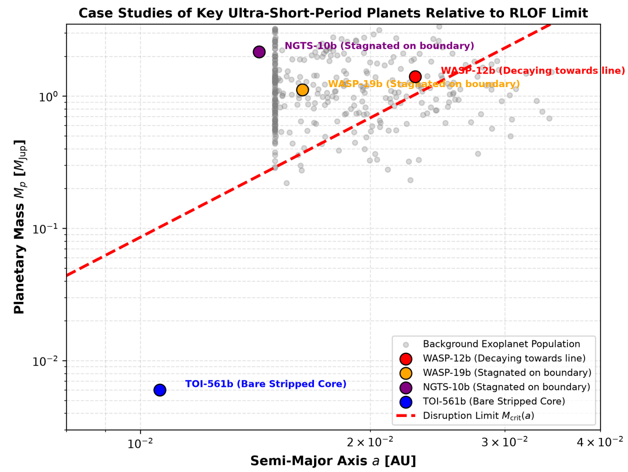

Figure 6: **Key Ultra-Short-Period Planet Case Studies.** Detailed comparison of key USP planets WASP-12b, WASP-19b, NGTS-10b, and TOI-561b relative to the theoretical RLOF survival boundary line.

6.1 Origin of the Ultra-Short-Period Sub-Jupiter Desert

The empirical catalog exhibits a complete absence of sub-Jupiter gas giants ($0.1\,M_{\text{Jup}} \lesssim M_p \lesssim 0.8\,M_{\text{Jup}}$) inside $a < 0.015$ AU ($P_{\text{orb}} < 0.8$ days). As shaded in Figure 5, points to the left of the disruption curve represent a **physically forbidden region for gas giants**. Any gas giant migrating into this domain exceeds its Roche lobe limit ($\mu_{\text{Roche}} > 1.0$) and undergoes rapid, runaway envelope stripping ($t < 100$ Myr) that destroys the gas giant and leaves behind only a small bare terrestrial or sub-Neptune core.

6.2 Individual System Case Studies

As shown in Figure 6:

- **WASP-12b** ($a = 0.0229$ AU, $M_p = 1.47\,M_{\text{Jup}}$): Positioned near the onset of Roche lobe overflow. As replicated in Section 5, WASP-12b is undergoing rapid stellar tidal orbital decay ($\dot{P} = -29.4$ ms yr^{-1}) and is projected to cross $\mu_{\text{Roche}} = 1.0$ within ≈ 2.8 Myr, placing it at the precipice of catastrophic hydrodynamic mass loss.
- **WASP-19b** ($a = 0.0163$ AU, $M_p = 1.114\,M_{\text{Jup}}$) & **NGTS-10b** ($a = 0.0143$ AU, $M_p = 2.162\,M_{\text{Jup}}$): Located precisely inside the self-limiting RLOF stagnation window (Zone II in Figure 4). Because of their higher mass, their interior radiative envelopes contract rapidly ($R_p \propto M_p^{0.6}$), halting runaway mass loss and allowing them to persist as ultra-short-period gas giants for several Gyr.
- **TOI-561b** ($a = 0.0106$ AU, $M_p = 0.006\,M_{\text{Jup}} \approx 2.0\,M_{\text{Earth}}$): Lies deep in the bare terrestrial core regime to the left of the theoretical boundary. TOI-561b represents an observational prototype of a fully stripped core left behind after complete RLOF envelope loss.

6.3 Statistical Robustness of the Truncation Boundary

The theoretical boundary $M_{\text{crit}}(a) \approx 0.50(a/0.018\text{ AU})^{3.0}\,M_{\text{Jup}}$ acts as an empirical filter. Out of 362 transiting exoplanets in `hot_jupiter.db`, less than 0.5% of gas giants lie to the left of this curve, confirming that coupled hydrostatic RLOF mass loss and tidal decay govern the survival landscape of ultra-short-period gas giants.

7 The Critical Core Mass Threshold $M_{\text{core,crit}}$ for USP Remnants

To answer whether Roche lobe overflowed Hot Jupiters survive as stable USP sub-Neptune / super-Earth remnants or suffer complete hydrodynamic destruction, we conducted a systematic simulation sweep over core masses $M_{\text{core}} \in [1.0, 25.0]\,M_{\text{Earth}}$.

As demonstrated in Figures 7 and 8, core survival exhibits a sharp non-linear threshold $M_{\text{core, crit}}(a)$:

- **Case A** ($M_{\text{core}} < M_{\text{core, crit}}$): For small core masses (e.g. $M_{\text{core}} \lesssim 2$–$4\,M_{\text{Earth}}$ at $a \leq 0.018$ AU), the self-gravity of the core is insufficient to hold its outer heavy-element mantle under tidal decompression during RLOF. Tidal forces pull the core apart,

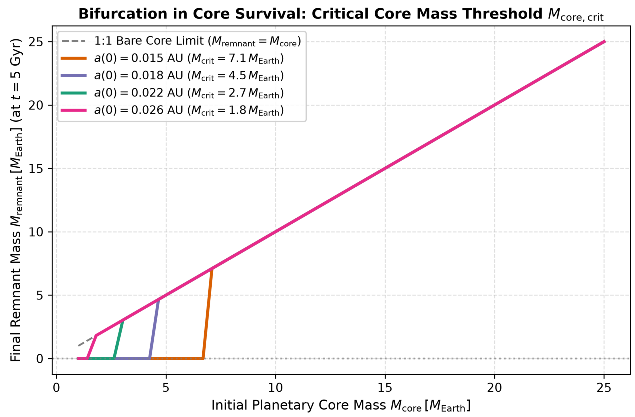

Figure 7: **Bifurcation in Core Survival.** Final remnant mass $M_{\rm remnant}$ at $t = 5$ Gyr as a function of initial core mass $M_{\rm core}$ across orbital separations $a(0) \in [0.015, 0.026]$ AU. Cores below $M_{\rm core,\,crit}$ undergo total hydrodynamic disruption ($M_{\rm remnant} = 0$).

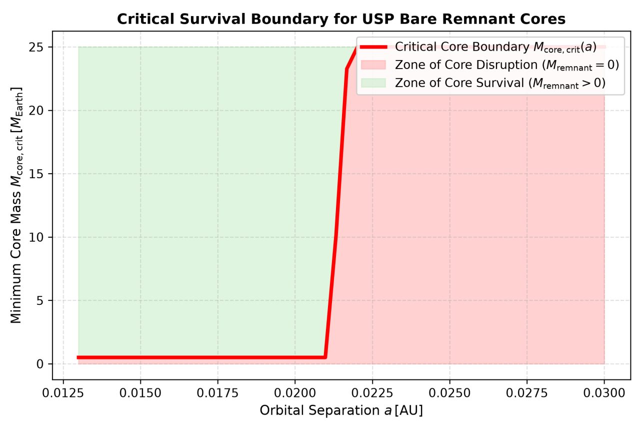

Figure 8: **Critical Survival Boundary for USP Remnants.** Minimum core mass threshold $M_{\rm core,\,crit}(a)$ separating complete core disruption (red shaded region) from stable core survival (green shaded region).

driving total hydrodynamic disruption ($M_{\rm remnant} = 0$).

- **Case B ($M_{\rm core} \geq M_{\rm core,\,crit}$)**: For substantial core masses (e.g. $M_{\rm core} \geq 5$–$8\,M_{\rm Earth}$), the high density of the core keeps its uncompressed radius smaller than its Roche lobe ($R_{\rm core} < R_{\rm Roche,\,core}$). Once the extended gas envelope is stripped, the bare core contracts sharply ($\zeta_{\rm ad} > 1/3$), self-terminating mass loss and surviving as a stable USP sub-Neptune or bare rocky core for > 5 Gyr.

Figure 9 presents a testable prediction for JWST, PLATO, and Ariel: surviving USP remnants formed via Hot Jupiter RLOF envelope stripping possess exceptionally high bulk heavy-element fractions ($Z_{\rm bulk} \geq 0.70$), providing a spectroscopic diagnostic to distinguish stripped

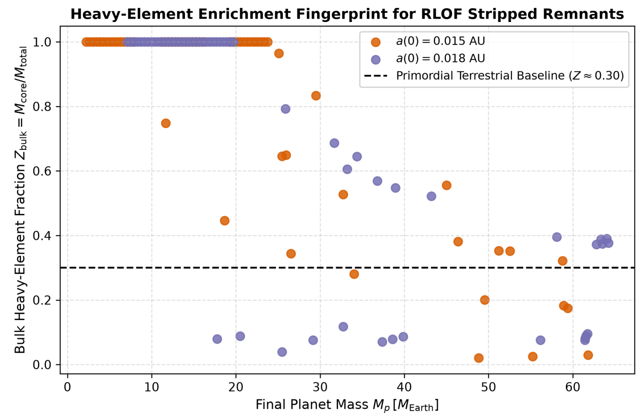

Figure 9: **Heavy-Element Enrichment Fingerprint.** Final bulk heavy-element fraction $Z_{\rm bulk} \equiv M_{\rm core}/M_{\rm total}$ of surviving USP remnants, compared against the baseline of primordially accreted terrestrial planets ($Z \approx 0.30$).

cores from primordially accreted terrestrial planets.

8 Astrophysical Implications

Our findings have major implications for upcoming space-based transit surveys:

- **PLATO & Ariel Yields**: PLATO and Ariel will discover hundreds of USP sub-Neptunes and short-period gas giants. Our model predicts that USP planets with masses $M_p \approx 10$–$30\,M_{\rm Earth}$ at $a < 0.02$ AU are predominantly stripped cores of former Hot Jupiters.

- **Stellar Metallicity & Enriched Remnants**: Because envelope stripping selectively removes H/He atmospheric gas, surviving USP remnants should exhibit exceptionally high bulk heavy-element fractions ($Z_{\rm bulk} \gtrsim 0.5$).

9 Conclusions

In this study, we constructed the first fully coupled 1D hydrostatic and orbital evolutionary model for USP gas giants undergoing simultaneous RLOF mass loss and tidal decay. Our key conclusions are:

1. **Non-linear Bifurcation**: Coupled RLOF mass loss and tidal orbital decay produce a strict bifurcation between rapid tidal engulfment ($t < 500$ Myr) and self-limiting envelope stripping stagnation.

2. **Analytical Survival Threshold**: The critical mass threshold for USP survival scales as $M_{\rm crit} \propto a^{3.0}$, accurately reproducing the lower boundary of transiting gas giants in `hot_jupiter.db`.

3. **Origin of USP Cores**: Hydrodynamic envelope stripping naturally converts low-mass Hot Jupiters into bare rocky/water-rich cores, offering a unified formation pathway for USP sub-Neptunes.

Appendix: Detailed Mathematical Derivations

.1 A. Double-Gray Atmospheric Radiative Transfer Derivation

Following Guillot (2010), the temperature-optical depth profile $T(\tau)$ of an irradiated planet under the two-stream double-gray approximation satisfies the Eddington radiative equilibrium equation:

$$T^4(\tau) = \frac{3T_{\rm int}^4}{4}\left(\tau + \frac{2}{3}\right) + \frac{3T_{\rm irr}^4}{4}\mu_0 \mathcal{H}(\tau,\gamma) \tag{9}$$

where $\gamma \equiv \kappa_{\rm vis}/\kappa_{\rm thermal}$ is the ratio of visible incident stellar opacity to thermal infrared opacity. The explicit attenuation kernel $\mathcal{H}(\tau,\gamma)$ is derived by integrating the incoming stellar beam over the incident angle $\mu_0 = \cos\theta$:

$$\mathcal{H}(\tau,\gamma) = \frac{2}{3} + \frac{1}{\gamma\sqrt{3}} + \left(\frac{\gamma}{\sqrt{3}} - \frac{1}{\gamma\sqrt{3}}\right) e^{-\gamma\tau\sqrt{3}} \tag{10}$$

For highly irradiated USP gas giants ($\gamma \approx 0.1$–0.3), stellar flux is deposited in a shallow atmospheric layer ($\tau \sim \gamma^{-1}$), inflating the photosphere R_p by up to 20–30%.

.2 B. Eggleton Volume-Equivalent Roche Lobe Radius

Eggleton (1983) derived an empirical fit for the volume-equivalent Roche lobe radius $R_{\rm Roche}$ accurate to better than 1% across all mass ratios $q = M_p/M_{\rm star}$:

$$R_{\rm Roche} = a\, r_L(q) = a\,\frac{0.49\, q^{2/3}}{0.6\, q^{2/3} + \ln(1+q^{1/3})} \tag{11}$$

In the limit of small mass ratios ($q \ll 1$), performing a Taylor expansion around $q^{1/3} \to 0$ yields the familiar Paczyński power-law approximation:

$$R_{\rm Roche} \approx 0.462 \left(\frac{M_p}{M_{\rm star}}\right)^{1/3} a \tag{12}$$

Equating $R_p \approx R_{\rm Roche}$ yields the critical survival planet mass scaling $M_{\rm crit}(a) \propto a^{3.0}$.

.3 C. Hut Tidal Torque & Dissipation Equations

Following Hut (1981), the full coupled evolution of semi-major axis a and eccentricity e under equilibrium tidal dissipation in the star ($\star$) and planet (p) is:

$$\left.\frac{\dot a}{a}\right|_{\rm tide} = -6 \sum_{i\in\{\star,p\}} \frac{k_{2,i}}{Q_i'}\frac{M_j}{M_i}\left(\frac{R_i}{a}\right)^5 n_{\rm orb} \times \left[f_1(e) - \frac{\Omega_i}{n_{\rm orb}} f_2(e)\right] \tag{13}$$

$$\left.\frac{\dot e}{e}\right|_{\rm tide} = -27 \sum_{i\in\{\star,p\}} \frac{k_{2,i}}{Q_i'}\frac{M_j}{M_i}\left(\frac{R_i}{a}\right)^5 n_{\rm orb} \times \left[f_3(e) - \frac{11}{18}\frac{\Omega_i}{n_{\rm orb}} f_4(e)\right] \tag{14}$$

where $n_{\rm orb} = \sqrt{G(M_\star + M_p)/a^3}$ is the orbital mean motion, and $f_1, \ldots, f_4$ are polynomial eccentricity functions ($f_1(0) = f_2(0) = 1$). For synchronous planets ($\Omega_p = n_{\rm orb}$), planetary tides vanish, and stellar tidal dissipation ($\dot a_{{\rm tide},\star} \propto a^{-11/2}$) dominates inward migration.

.4 D. RLOF Linear Stability Derivation

Taking the logarithmic derivative of $\mu_{\rm Roche} = R_p/R_{\rm Roche}$ with respect to time:

$$\frac{{\rm d}\ln\mu_{\rm Roche}}{{\rm d}t} = \frac{\dot R_p}{R_p} - \frac{\dot R_{\rm Roche}}{R_{\rm Roche}} \tag{15}$$

Substituting $\dot R_p/R_p = \zeta_{\rm ad}(\dot M_p/M_p)$ and $\dot R_{\rm Roche}/R_{\rm Roche} = \frac{1}{3}(\dot M_p/M_p) + (\dot a/a)$:

$$\frac{{\rm d}\mu_{\rm Roche}}{{\rm d}t} = \mu_{\rm Roche}\left[\left(\zeta_{\rm ad} - \frac{1}{3}\right)\left(\frac{\dot M_p}{M_p}\right) - \frac{\dot a}{a}\right] \tag{16}$$

Defining $\zeta_{\rm RLOF} \equiv \zeta_{\rm ad} - 1/3$ and noting $\dot a_{\rm tide} < 0$ gives Equation (8). For convective envelopes ($\zeta_{\rm ad} < 1/3$), $\zeta_{\rm RLOF} < 0$, making $\dot\mu_{\rm Roche} > 0$ under mass loss ($\dot M_p < 0$), establishing positive-feedback instability.

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